Natural Deduction 3

Dominik Kirst

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Recap

Last week, we studied semantics of ND and completeness theorems:

- Henkin's completeness proof
- Constructive refinement of Henkin's completeness proof
- Kripke semantics for intuitionistic ND
- Kripke completeness as semantic cut-elimination

Plan for Today

- Algebraic semantics:
  - Complete Heyting and Boolean algebras
  - Completeness proofs
- Arithmetical systems:
  - Heyting arithmetic and Peano arithmetic
  - Some meta-theoretical properties
- Friedman's A-translation:
  - Syntactic definition and correctness
  - Applications

First part still based on this paper:
`https://www.ps.uni-saarland.de/Publications/documents/ForsterEtAl_2021_Completeness.pdf`

Heyting and Boolean Algebras

Previous semantics mostly interpreted logical connectives in the meta-theory:

$$\mathcal{M} \vDash A \wedge B \qquad := \qquad \mathcal{M} \vDash A \text{ “and” } \mathcal{M} \vDash B$$

This can be generalised to arbitrary algebraic structures capturing the same behaviour:

Definition

We call $\mathcal{A} = (\Omega, \leq, 0, 1, \sqcap, \sqcup, \sqsupset)$ a **Heyting algebra** if:

- $\leq$ is a preorder on Ω with 0 being minimal and 1 being maximal
- $\sqcap$ forms greatest lower bounds / meets: $z \leq x \sqcap y$ if and only if $z \leq x$ and $z \leq y$
- $\sqcup$ forms least upper bounds / joins: $x \sqcup y \leq z$ if and only if $x \leq z$ and $y \leq z$
- $\sqsupset$ forms Heyting implication: $z \leq x \sqsupset y$ if and only if $z \sqcap x \leq y$

We call $\mathcal{A}$ a **Boolean algebra** if moreover $1 \leq x \sqcup (x \sqsupset 0)$.

This generalises previous semantics as the meta-level truth values form such algebras!

Soundness

Given any Heyting algebra $\mathcal{A}$ and a valuation $V(P) \in \Omega$ for each atom define:

$$\begin{aligned} [\![P]\!] &:= V(P) & [\![A \wedge B]\!] &:= [\![A]\!] \sqcap [\![B]\!] \\ [\![\bot]\!] &:= 0 & [\![A \vee B]\!] &:= [\![A]\!] \sqcup [\![B]\!] \\ [\![\top]\!] &:= 1 & [\![A \Rightarrow B]\!] &:= [\![A]\!] \sqcup [\![B]\!] \end{aligned}$$

Theorem (Soundness)

If $\vdash A$ (or $\vdash_c A$), then $1 \leq [\![A]\!]$ in any Heyting (or Boolean) algebra.

Proof.

Show by induction on a derivation $\Gamma \vdash A$ that $[\![\bigwedge \Gamma]\!] \leq [\![A]\!]$. □

Corollary

In NJ the formula $A \vee \neg A$ is not derivable.

Proof.

Consider the linearly-ordered three-valued Heyting algebra $(0, \star, 1)$. □

Completeness

The completeness for algebraic semantics follows from a simple observation:

Fact (Lindenbaum Algebra)

The formulas preorderd by $A \vdash B$ (or $A \vdash_c B$) form a Heyting (or Boolean) algebra.

Proof.

- Reflexivity by Ax rule, transitivity by IE rule, minimality of $\bot$ by Exp.
- The equivalences for $\sqcap$, $\sqcup$ and $\sqsupset$ reduce to simple ND derivations. □

Corollary (Completeness)

If $1 \leq [\![A]\!]$ in any Heyting (or Boolean) algebra, then $\vdash A$ (or $\vdash_c A$).

Proof.

In the respective Lindenbaum algebra $1 \leq [\![A]\!]$ immediately yields derivability of A. □

Complete Algebras

We extend the algebraic semantics to first-order quantifiers:

Definition

A Heyting algebra $\mathcal{A}$ is called **complete** if it has arbitrary greatest lower bounds

$$y \leq \bigsqcap S \text{ if and only if } y \leq x \text{ for all } y \in S$$

and then $\mathcal{A}$ also has arbitrary least upper bounds $\bigsqcup S := \bigsqcap\{x \mid \forall y \in S.\, y \leq x\}$.

Given a complete Heyting algebra, we can interpret quantifiers in a natural way:

$$[\![\forall x A]\!] := \bigsqcap_t [\![A[t/x]]\!] \qquad [\![\exists x A]\!] := \bigsqcup_t [\![A[t/x]]\!]$$

- Soundness: show by induction on $\Gamma \vdash A$ that $[\![\bigwedge \Gamma[\sigma]]\!] \leq [\![A[\sigma]]\!]$
- Completeness: needs more work as the Lindenbaum algebras are not complete

Mac-Neille Completion

Luckily, there is a general recipe how algebras can be completed!

Fact

Every Heyting algebra $\mathcal{A}$ embeds into a complete Heyting algebra $\overline{\mathcal{A}}$.

Proof.

1. Define $L(S) := \{x \in \Omega \mid \forall y \in S.\, x \leq y\}$ and $U(S) := \{x \in \Omega \mid \forall y \in S.\, y \leq x\}$.
2. Consider the set $\overline{\Omega} := \{S \subseteq \Omega \mid L(U(S)) \subseteq S\}$ ordered by inclusion.
3. Equip $\overline{A}$ with algebraic structure as follows:

$$\overline{0} := \{x \in \Omega \mid x \leq 0\} \qquad S \overline{\sqcap} S' := S \cap S'$$
$$S \overline{\sqsupset} S' := \{x \in \Omega \mid \forall y \in S.\, x \sqcup y \in S'\} \qquad S \overline{\sqcup} S' := L(U(S \cup S'))$$

4. Observe that $\overline{\Omega}$ is complete by using intersection $\bigcap$ (and union $\bigcup$).
5. Verify that $\overline{x} := \{y \in \Omega \mid y \leq x\}$ embeds $\mathcal{A}$ into $\overline{\mathcal{A}}$. □

Completeness

Theorem (Completeness)

If $1 \leq [\![A]\!]$ *in any complete Heyting (or Boolean) algebra, then* $\vdash A$ *(or* $\vdash_c A$*).*

Proof.

1. Assume that $\overline{1} \subseteq [\![A]\!]$ in the completion $\overline{\mathcal{L}}$ of the Lindenbaum algebra for $\vdash A$.
2. Verify that evaluation in $\overline{\mathcal{L}}$ coincides with the embedding of $\mathcal{L}$: $[\![A]\!] = \overline{A}$.
3. But then in particular $\overline{1} \subseteq \overline{A}$ and hence $1 \leq A$, meaning $\vdash A$.
4. To cover $\vdash_c A$, observe that the completion of Boolean algebras is Boolean. □

Note that all algebraic completeness proofs as presented here are constructive!

Heyting Arithmetic and Peano Arithmetic

In the language of arithmetics $(0, S, +, \cdot)$ with equality consider the following axioms

$$x + 0 = x \qquad x + (Sy) = S(x + y)$$
$$x \cdot 0 = 0 \qquad x \cdot (Sy) = x + x \cdot y$$
$$0 \neq Sx \qquad x = y \text{ whenever } Sx = Sy$$

as well as the following axiom scheme for every formula A:

$$A(0) \Rightarrow (\forall x A(x) \Rightarrow A(Sx)) \Rightarrow \forall x A(x)$$

Definition

Heyting arithmetic (HA) consists of the above axioms and is based on the system NJ, Peano arithmetic (PA) consists of the same axioms but is based on NK.

The Standard Model

Definition

We refer to the set $\mathbb{N}$ of natural numbers equipped with the arithmetical operations as the **standard model** of HA and PA.

Fact

$\mathbb{N}$ satisfies all axioms of HA, so whenever $HA \vdash A$ also $\mathbb{N} \vDash A$.

Due to the compactness theorem, there are also many non-standard models:

1. Extend the language with a constant c and consider axioms $\overline{n} \leq c$ for all $n \in \mathbb{N}$.
2. Every finite set Γ of these axioms is satisfied by $\mathbb{N}$ by choosing c large enough.
3. Then by compactness there is a model $\mathcal{M}$ satisfying all these axioms at once.
4. Observe that $\mathcal{M}$ differs from $\mathbb{N}$ as c in is larger than every standard number $\overline{n}$.

Meta-Theoretic Properties

Some standard (but potentially confusing) meta-theoretic results are:

- Completeness: if $\mathcal{M} \vDash A$ for all models $\mathcal{M}$ of PA, then $PA \vdash_c A$
- Incompleteness 1: if $\mathbb{N} \vDash A$, then not necessarily $PA \vdash_c A$
- Incompleteness $1'$: there are sentences A with neither $PA \vdash_c A$ nor $PA \vdash_c \neg A$
- Incompleteness 2: $PA \nvdash_c Con(PA)$ expressing consistency of PA

Constructively, there are some even more interesting phenomena:

- $\mathbb{N}$ is the only model of HA, a consequence of Tennenbaum's theorem
- PA does not have a model at all, while it is still provably consistent

Friedman's A-Translation: Definition

Observe that the double-negation translation A^* can be verified in minimal logic:

$$\vdash_c A \qquad \text{if and only if} \qquad \vdash_m A^*$$

Therefore we can generalise the translation to an arbitrary formula $A_\bot$

$$P^{A_\bot} := \sim\sim P \qquad (A \vee B)^{A_\bot} := \sim\sim(A^{A_\bot} \vee B^{A_\bot})$$
$$(A \Rightarrow B)^{A_\bot} := A^{A_\bot} \Rightarrow B^{A_\bot} \qquad (\forall x A)^{A_\bot} := \forall x A^{A_\bot}$$
$$(A \wedge B)^{A_\bot} := A^{A_\bot} \wedge B^{A_\bot} \qquad (\exists x A)^{A_\bot} := \sim\sim\exists x A^{A_\bot}$$

where $\sim A$ is a shorthand for $A \Rightarrow A_\bot$.

Theorem (Friedman)

For arbitrary A and $A_\bot$, we have that $\vdash_c A$ implies $\vdash_m A^{A_\bot}$.

Friedman's A-Translation: Proof Outline

1. Double-negation elimination holds for translated formulas: $\vdash_m \sim\sim A^{A_\bot} \Rightarrow A^{A_\bot}$
   - By induction on A, exploiting that minimal logic identifies $\sim\sim\sim B$ and $\sim B$

2. Derivations can be replayed in the translation: $\Gamma \vdash_c A$ implies $\Gamma^{A_\bot} \vdash_m A^{A_\bot}$
   - By induction on $\Gamma \vdash_c A$, using (1) whenever negative reasoning is needed

Note however that it is **not** the case that classically A and $A^{A_\bot}$ are always equivalent: If we choose $A_\bot := \top$ then we have $\vdash_c \bot^{A_\bot}$ but not $\vdash_c \bot$!

Partial Conservativity Results

The A-Translation can be used to show that some derivations of PA work also in HA!

Theorem

Let $t(x)$ be given and assume $HA \vdash \neg\neg\exists x\, t(x) = 0$. Then $HA \vdash \exists x\, t(x) = 0$.

Proof.

1. $HA \vdash \neg\neg\exists x\, t(x) = 0$ in particular $PA \vdash_c \exists x\, t(x) = 0$.
2. Choosing $A_\perp := \exists x\, t(x) = 0$, we have $PA^{A_\perp} \vdash (\exists x\, t(x) = 0)^{A_\perp}$.
3. We can derive $HA \vdash PA^{A_\perp}$, so we have $HA \vdash (\exists x\, t(x) = 0)^{A_\perp}$.
4. From this we can derive $HA \vdash \exists x\, t(x) = 0$ as desired. □

More generally for sentences A of the form $\forall x \exists y\, B(x, y)$ with B quantifier-free:

- If A is provable in PA, then A is already provable in HA
- If A is provable in ZF, then A is provable in IZF (intuitionistic Zermelo-Fraenkel)